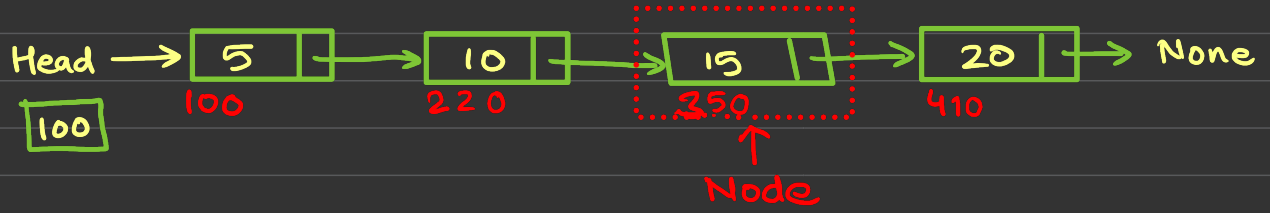
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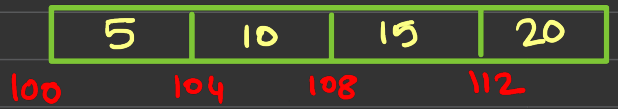
Introduction to Linked List

→ A Linked List is a linear data structure where elements (nodes) are stored in separate memory locations and are connected to each other.

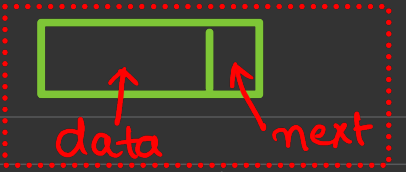
Example :- Store a list of numbers [5, 10, 15, 20] in Linked List



→ Difference between Arrays and Linked List ?



→ Linked List → Collection of **Nodes**



→ Node contains **data** and pointer to the next node.

→ In Linked List, nodes can contain

① Homogeneous data





② Heterogeneous data



③ Record Data

→ Student Record data → Student(Name, Rollno, Marks)



→ Linked Lists has following properties

① Successive nodes are connected by pointers

② Last element points to None

③ It can be made just as long as required

④ Head points to the first node.

→ Operations on a Linked List

Operation	Time Complexity
Insert at beginning	O(1)
Insert at Position (itm)	O(n)
Delete at beginning	O(1)
Delete at Position (itm)	O(n)
Search node	O(n)
Display all items	O(n)

→ Insertion in Linked List

Head → None

① Insert(10)

→ Create an empty node (temp)
→ Add value(10) and next(None)
→ Make head point to temp



② Insert(5,0) ← Insert in the beginning.

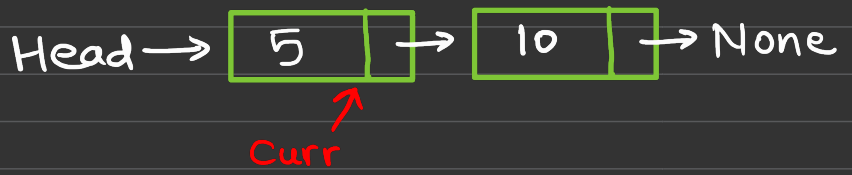
→ Create an empty node (temp)
→ Add value(5) and next(Head)
→ Make head point to temp



③ Insert(2, 1) ← Insert in between

→ Init curr = Head.

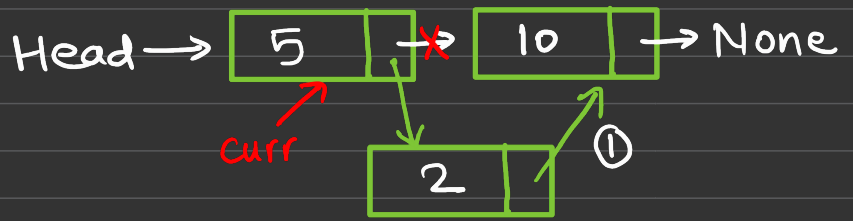
→ Iterate curr such that it points (i-1)th node.



→ Create an empty node (temp)

→ Add value(2) and next(curr.next)

→ Make Curr.next point to the temp.



④ Insert(6,1) ← Insert in the end

→ Init curr = Head.

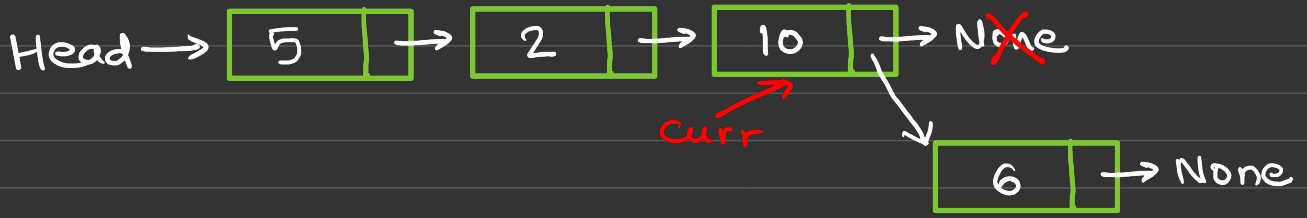
→ Iterate curr such that it points to the last node



→ Create an empty node (temp)

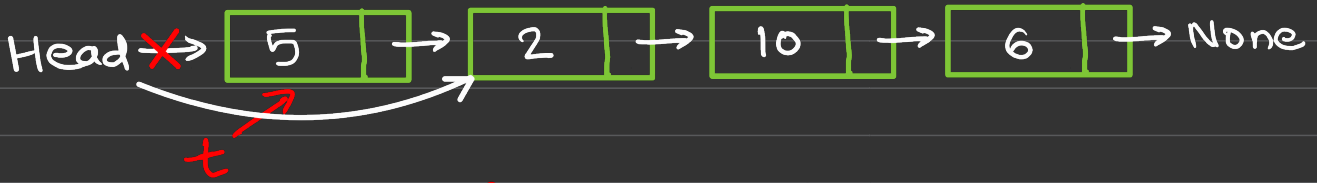
→ Add value(6) and next(None)

→ Make curr.next point to the temp.



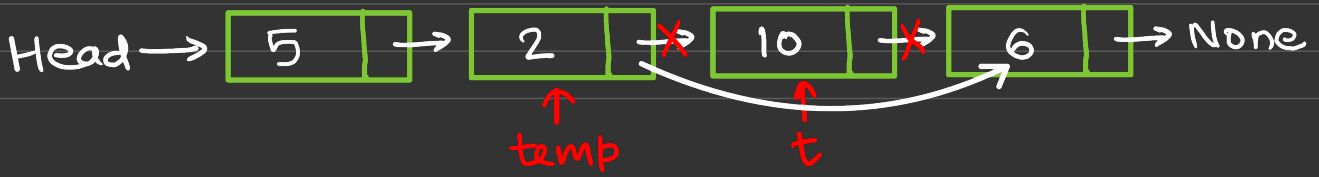
⑤ Delete(0) ← Delete in the beginning.

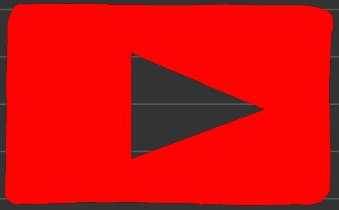
→ Create a temp var(t) which references head.
→ make head point to head.next
→ free memory pointed by t



⑥ Delete(2) ← Delete in between/end

→ make temp = head. and iterate temp to point (i-1)th node
→ make t = temp.next (node to be deleted)
→ mark temp.next = t.next
→ free the node t



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